

Multiple Choice

1. **Answer:** A

Options: A) 17 O; B) 19 F; C) 29 Si; D) 31 P

Note: Correct option: A - 17 O

2. **Answer:** A

Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays

Note: Correct option: A - Neutrons

3. **Answer:** D

Options: A) I only; B) II only; C) III only; D) I and III only

Note: Correct option: D - I and III only

4. **Answer:** C

Options: A) 0.5 T; B) 1.2 T; C) 2.9 T; D) 100 T

Note: Correct option: C - 2.9 T

5. **Answer:** B

Options: A) 0.0471 Hz; B) 21.2 Hz; C) 42.4 Hz; D) 66.7 Hz

Note: Correct option: B - 21.2 Hz

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '345.0 MHz'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' H_3O^+ / H_2O '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It is a red-brown volatile liquid.'

Long Form Response

11. **Answer:** It has little effect.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** MgO. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.